```
clc; clear; close all;

f = 8:0.01:12;
Z0 = 50; c = 3e8;
beta = 2*pi*f*1e9/c;

% S-parameters (illustrative model)
S11 = -25-15*(f-10).^2-35*exp(-((f-9)/0.1).^2) ...
    -220*exp(-((f-10)/0.08).^2);
S11 = max(S11,-240);
S22 = max(S11+2,-240);
S21 = -0.025-0.055*((f-10)/2).^2;
S12 = S21;
RL = -S11; IL = -S21;
Eff = 99.4-1.2*((f-10)/2).^2;
Zc = 50+1.25*(f-10).^2;

figure('Color','w');
Y = {S11,S21,S12,S22,RL,IL,Eff,beta,Zc};
T = {'Input Reflection S11','Forward Transmission S21',...
    'Reverse Transmission S12','Output Reflection S22',...
    'Return Loss','Insertion Loss','Transmission Efficiency',...
    'Propagation Constant','Characteristic Impedance'};
C = {'b','r','m','k','b','r','g',[0.5 0 0.8],[0.9 0.4 0.1]};

for k = 1:9
    subplot(3,3,k);
    plot(f,Y{k},'Color',C{k},'LineWidth',1.5);
    title(T{k}); xlabel('Frequency (GHz)'); grid on;
    xlim([8 12]);
end

ylabel('Value');
sgtitle('Microwave Two-Port Network: 8-12 GHz');
```

# Microwave Two-Port Network: 8-12 GHz

